

Лабораторная работа №8

Планировщики событий

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Цель работы

Получить навыки работы с планировщиками задач **cron** и **at** в операционной системе Linux.

Ход выполнения работы

Проверка службы crond

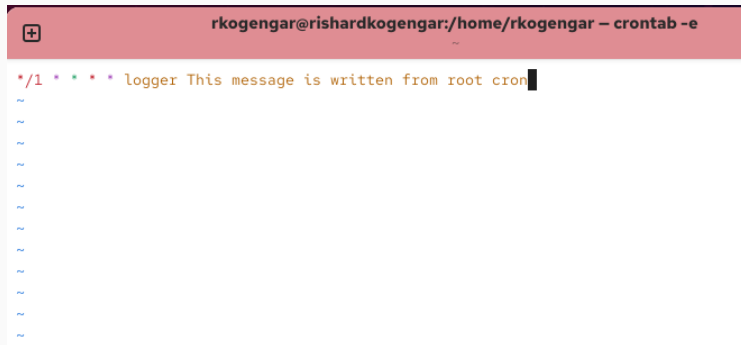
```
root@rishardkogengar:/home/rkogengar# systemctl status crond -l
● crond.service - Command Scheduler
   Loaded: loaded (/usr/lib/systemd/system/crond.service; enabled; preset: enabled)
   Active: active (running) since Sun 2025-10-05 19:23:16 MSK; 3min 4s ago
 Invocation: 5543b9829ab04d0a84a75277b6f4ff22
    Main PID: 1203 (crond)
      Tasks: 1 (limit: 24779)
     Memory: 1M (peak: 1.1M)
        CPU: 5ms
    CGroup: /system.slice/crond.service
            └─1203 /usr/sbin/crond -n

Oct 05 19:23:16 rishardkogengar.localdomain systemd[1]: Started crond.service - Command Scheduler.
Oct 05 19:23:16 rishardkogengar.localdomain crond[1203]: (CRON) STARTUP (1.7.0)
Oct 05 19:23:16 rishardkogengar.localdomain crond[1203]: (CRON) INFO (Syslog will be used instead)
Oct 05 19:23:16 rishardkogengar.localdomain crond[1203]: (CRON) INFO (RANDOM_DELAY will be scaled)
Oct 05 19:23:16 rishardkogengar.localdomain crond[1203]: (CRON) INFO (running with inotify support)
root@rishardkogengar:/home/rkogengar# cat /etc/crontab
SHELL=/bin/bash
PATH=/sbin:/bin:/usr/sbin:/usr/bin
MAILTO=root

# For details see man 4 crontabs

# Example of job definition:
# .----- minute (0 - 59)
# | .----- hour (0 - 23)
# | | .----- day of month (1 - 31)
# | | | .----- month (1 - 12) OR jan,feb,mar,apr ...
# | | | | .---- day of week (0 - 6) (Sunday=0 or 7) OR sun,mon,tue,wed,thu,fri,sat
# | | | | |
# * * * * * user-name  command to be executed

root@rishardkogengar:/home/rkogengar# crontab -l
no crontab for root
root@rishardkogengar:/home/rkogengar#
```



A terminal window titled "rkogengar@rishardkogengar:/home/rkogengar – crontab -e". The window shows a crontab editor interface with a red header bar. The main area is white with a red prompt character. The first line of the crontab is "* /1 * * * * logger This message is written from root cron", where the asterisks are colored (red, green, blue, red, blue, red) and the rest is black. A black cursor is at the end of the line. Below the first line, there are several empty lines, each preceded by a red prompt character.

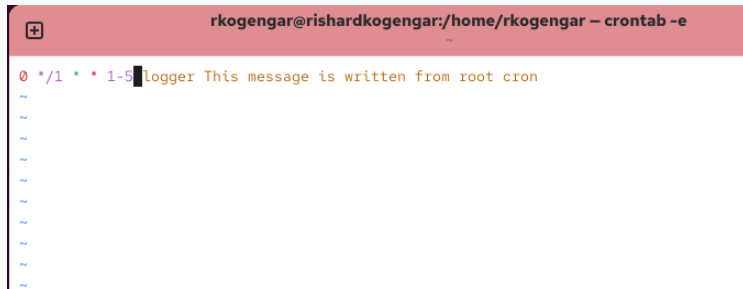
```
rkogengar@rishardkogengar:/home/rkogengar – crontab -e
*/1 * * * * logger This message is written from root cron
~
~
~
~
~
~
~
~
~
~
~
```

Рис. 2: Редактирование crontab для root

Проверка выполнения задания

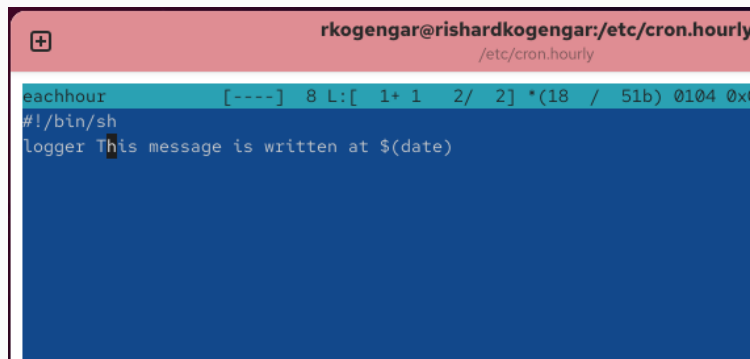
```
root@rishardkogengar:/home/rkogengar# crontab -e
no crontab for root - using an empty one
crontab: installing new crontab
root@rishardkogengar:/home/rkogengar# crontab -l
*/1 * * * * logger This message is written from root cron
root@rishardkogengar:/home/rkogengar# grep written /var/log/messages
Oct  5 19:30:01 rishardkogengar root[4264]: This message is written from root cron
root@rishardkogengar:/home/rkogengar# grep written /var/log/messages
Oct  5 19:30:01 rishardkogengar root[4264]: This message is written from root cron
Oct  5 19:31:01 rishardkogengar root[4439]: This message is written from root cron
root@rishardkogengar:/home/rkogengar# grep written /var/log/messages
Oct  5 19:30:01 rishardkogengar root[4264]: This message is written from root cron
Oct  5 19:31:01 rishardkogengar root[4439]: This message is written from root cron
Oct  5 19:32:01 rishardkogengar root[4607]: This message is written from root cron
root@rishardkogengar:/home/rkogengar#
```

Рис. 3: Результат выполнения cron-задания



```
rkogengar@rishardkogengar:/home/rkogengar - crontab -e
0 */1 * * 1-5 logger This message is written from root cron
~
~
~
~
~
~
~
~
~
~
```

Рис. 4: Изменение расписания cron



The screenshot shows a terminal window with a pink title bar. The title bar contains a plus icon, the username 'rkogengar@rishardkogengar', and the current directory path '/etc/cron.hourly'. Below the title bar, the terminal content is as follows:

```
eachhour      [----] 8 L:[ 1+ 1 2/ 2] *(18 / 51b) 0104 0x
#!/bin/sh
logger This message is written at $(date)
```

Рис. 5: Создание сценария eachhour

Файл расписания eachhour в /etc/cron.d

```
rkogengar@rishardkogengar:/etc/cron.d - mcedit eachhour
/etc/cron.d

eachhour [-----] 63 L:[ 1+ 0 1/ 1] *(63 / 63b) <EOF>
11 * * * * root logger This message is written from /etc/cron.d
```

Проверка службы atd

```
root@rishardkogengar:/etc/cron.d# systemctl status atd
● atd.service - Deferred execution scheduler
   Loaded: loaded (/usr/lib/systemd/system/atd.service; enabled; preset: enabled)
   Active: active (running) since Sun 2025-10-05 19:23:16 MSK; 14min ago
     Invocation: 78d40913749e4b23bd64206d10e7c50a
       Docs: man:atd(8)
    Main PID: 1201 (atd)
      Tasks: 1 (limit: 24779)
     Memory: 216K (peak: 1.1M)
```

Заключение

В ходе работы были изучены инструменты планирования задач **cron** и **at**.

Были освоены принципы создания расписаний, настройки периодичности и автоматизации системных действий.

Полученные знания позволяют эффективно управлять периодическими и одноразовыми задачами в Linux.